

Ying Zheng, NY, HW1.

Prob 1

(a). $\{z_t\}_{t=1}^{\infty}$ is iid $\sim N(0, \sigma^2)$

$$\therefore E(z_t) = 0, \text{Cov}(z_t, z_s) = \begin{cases} \sigma^2, & t=s \\ 0, & t \neq s. \end{cases}$$

$$E(z_t^2) = \text{Var}(z_t) + E(z_t)^2 = \sigma^2 + 0 = \sigma^2, \text{Var}(z_t) = \sigma^2.$$

when $x_t = a + b z_t + c z_{t-1}$

$$E(x_t) = E(a) + b E(z_t) + c E(z_{t-1}) = 0.$$

$$\begin{aligned} \gamma(t+h, t) &= \text{Cov}(x_{t+h}, x_t) \\ &= \text{Cov}(a + b z_{t+h} + c z_{t+h-1}, a + b z_t + c z_{t-1}) \\ &= b^2 \text{Cov}(z_{t+h}, z_t) + b c \text{Cov}(z_{t+h}, z_{t-1}) + b c \text{Cov}(z_{t+h-1}, z_t) + c^2 \text{Cov}(z_{t+h-1}, z_{t-1}) \\ &= \begin{cases} \sigma^2(b^2 + c^2), & h=0 \\ \sigma^2 b c, & h=\pm 1 \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

$\therefore \gamma(t+h, t)$ does not depend on t $\therefore \{x_t\}$ is stationary $\square$.

(b) when $x_t = a + z_t^2$

$$E(x_t) = E(z_t^2) = \sigma^2 + a.$$

$$\begin{aligned} \gamma(t+h, t) &= \text{Cov}(x_{t+h}, x_t) \\ &= \text{Cov}(a + z_{t+h}^2, a + z_t^2) \\ &= E[z_{t+h}^2 z_t^2] - \sigma^4 = \begin{cases} 2\sigma^4, & h=0 \\ 0, & h \neq 0 \end{cases} \end{aligned}$$

$\therefore \gamma(t+h, t)$ does not depend on t $\therefore \{x_t\}$ is stationary $\square$.

(c) when $x_t = z_t z_{t-1}$

$$E(x_t) = E(z_t z_{t-1}) = E(z_t) E(z_{t-1}) = 0$$

$$\begin{aligned} \gamma(t+h, t) &= \text{Cov}(z_{t+h} z_{t+h-1}, z_t z_{t-1}) \\ &= E[z_{t+h} z_{t+h-1} z_t z_{t-1}] - E[z_{t+h} z_{t+h-1}] E[z_t z_{t-1}] \\ &= \begin{cases} E[z_t^2 z_{t-1}^2] = E[z_t^2] E[z_{t-1}^2] = \sigma^4, & h=0 \\ E[z_t z_{t-1} z_{t+1} z_t] = 0, & h=1 \\ E[z_{t+1} z_t^2 z_{t+1}] = 0, & h=-1 \\ 0, & \text{otherwise.} \end{cases} = \begin{cases} \sigma^2, & h=0 \\ 0, & \text{otherwise} \end{cases} \therefore \{x_t\} \text{ stationary } \square. \end{aligned}$$

Prob 2.

$$\{\varepsilon_t\} \sim iid \sim WN(0, \sigma^2), \quad E[\varepsilon_t^2] = \sigma^2.$$

$$\text{when } x_t = \varepsilon_t + \varepsilon_{t-1}\varepsilon_{t-2}$$

$$E(x_t) = E(\varepsilon_t) + E(\varepsilon_{t-1})E(\varepsilon_{t-2}) = 0$$

$$\gamma_x(h, t) = \text{cov}(x_{t+h}, x_t)$$

$$= \text{cov}(\varepsilon_{t+h} + \varepsilon_{t+h-1}\varepsilon_{t+h-2}, \varepsilon_t + \varepsilon_{t-1}\varepsilon_{t-2})$$

$$= E[(\varepsilon_{t+h} + \varepsilon_{t+h-1}\varepsilon_{t+h-2})(\varepsilon_t + \varepsilon_{t-1}\varepsilon_{t-2})] - E[\varepsilon_{t+h} + \varepsilon_{t+h-1}\varepsilon_{t+h-2}]E[\varepsilon_t + \varepsilon_{t-1}\varepsilon_{t-2}]$$

$$= E[\varepsilon_{t+h}\varepsilon_t] + E[\varepsilon_{t+h}\varepsilon_{t-1}\varepsilon_{t-2}] + E[\varepsilon_{t+h-1}\varepsilon_{t+h-2}\varepsilon_t] + E[\varepsilon_{t+h-1}\varepsilon_{t+h-2}\varepsilon_{t-1}\varepsilon_{t-2}]$$

$$= \begin{cases} E[\varepsilon_t^2] + E[\varepsilon_{t-1}^2\varepsilon_{t-2}^2] = \sigma^2 + \sigma^4, & h=0. \\ 0, & \text{otherwise.} \end{cases}$$

$\therefore x_t$ is weak white noise $\sim WN(0, \sigma^2 + \sigma^4)$

$$E[x_t | \mathcal{H}_{t-1}] = E[\varepsilon_t + \varepsilon_{t-1}\varepsilon_{t-2} | \mathcal{H}_{t-1}] = E[\varepsilon_t | \mathcal{H}_{t-1}] + E[\varepsilon_{t-1}\varepsilon_{t-2} | \mathcal{H}_{t-1}] = \varepsilon_{t-1}\varepsilon_{t-2} \neq 0.$$

$\therefore \{x_t\}$ is not MDS $\square$.

Prob 3.

$$E[y_t] = 0.$$

$$\text{when } x_t = y_t - y_{t-1},$$

$$E[x_t] = E[y_t] - E[y_{t-1}] = 0$$

$$\gamma_x(h, t) = \text{cov}(x_{t+h}, x_t)$$

$$= \text{cov}(y_{t+h} - y_{t+h-1}, y_t - y_{t-1})$$

$$= E[y_{t+h}y_t - y_{t+h}y_{t-1} - y_{t+h-1}y_t + y_{t+h-1}y_{t-1}]$$

$$= \gamma_y(h) - \gamma_y(h+1) - \gamma_y(h-1) + \gamma_y(h)$$

$$= 2\gamma_y(h) - \gamma_y(h+1) - \gamma_y(h-1) \quad \perp t.$$

$\therefore \{x_t\}$ stationary $\square$.

Data Processing

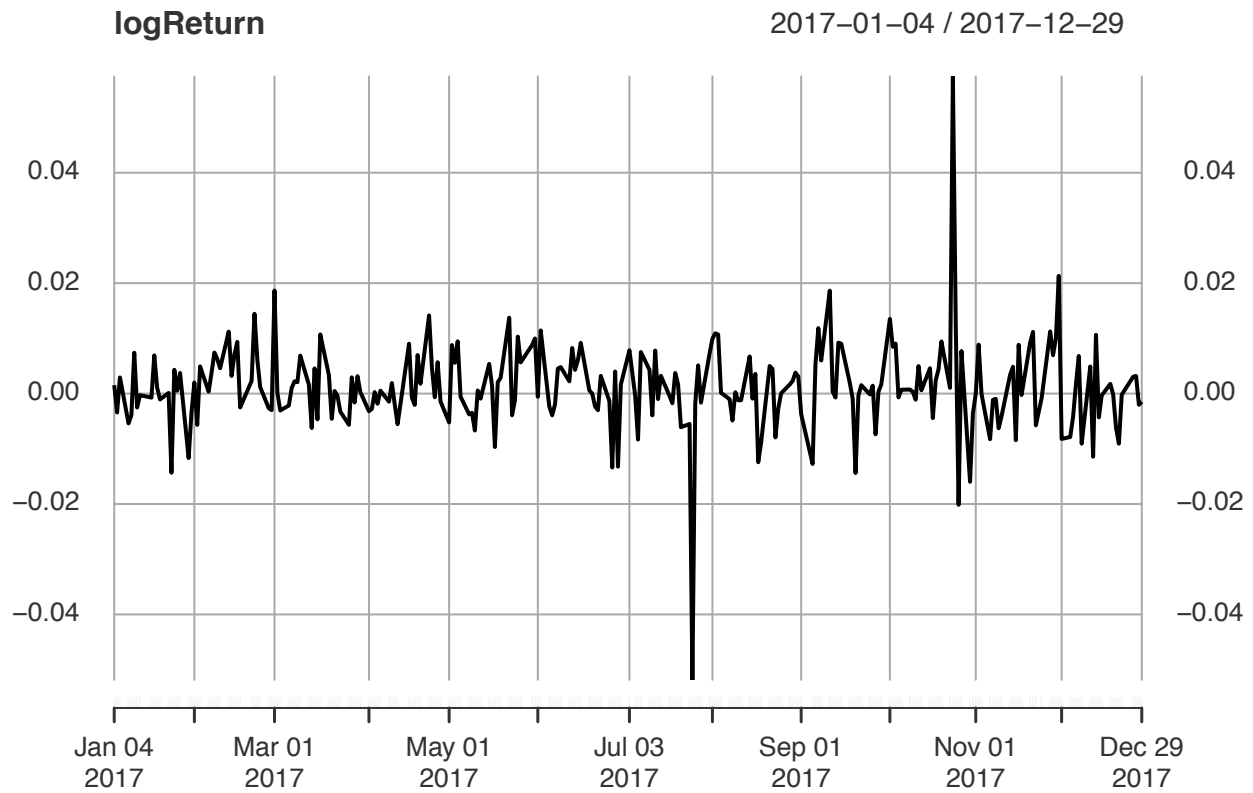
```
library('quantmod')

## Loading required package: xts
## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
## Loading required package: TTR
## Version 0.4-0 included new data defaults. See ?getSymbols.
adj = getSymbols("MMM", from="2017-1-1", to="2017-12-31", auto.assign=F)[,6]

## 'getSymbols' currently uses auto.assign=TRUE by default, but will
## use auto.assign=FALSE in 0.5-0. You will still be able to use
## 'loadSymbols' to automatically load data. getOption("getSymbols.env")
## and getOption("getSymbols.auto.assign") will still be checked for
## alternate defaults.
##
## This message is shown once per session and may be disabled by setting
## options("getSymbols.warning4.0"=FALSE). See ?getSymbols for details.
##
## WARNING: There have been significant changes to Yahoo Finance data.
## Please see the Warning section of '?getSymbols.yahoo' for details.
##
## This message is shown once per session and may be disabled by setting
## options("getSymbols.yahoo.warning"=FALSE).

Problem 4

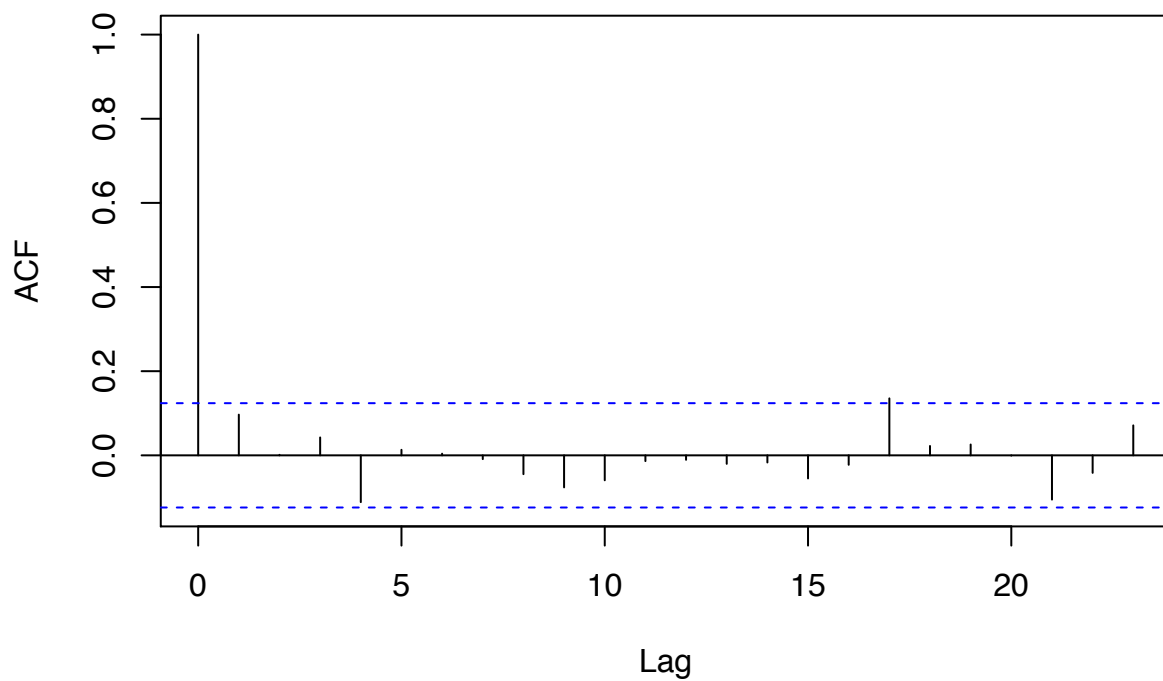
#(a)
logReturn = diff(log(adj))[-1]
plot(logReturn)
```



Non-stationarity is not obvious from this plot.

```
##(b)  
acf(logReturn)
```

Series logReturn



```
##(c)
Box.test(logReturn, lag = 20, type = "Ljung-Box")

##
## Box-Ljung test
##
## data: logReturn
## X-squared = 15.52, df = 20, p-value = 0.7459
p-value of 0.7459 suggests significant evidence of correlation.
```